



Department of Mechanical Engineering

Dear Fellowship Review Panel,

It is my pleasure to write in support of Mr. Jacob Vaught's application for the NSF Graduate Research Fellowship. I am a Professor of Computer Science and have supervised Jacob's arm of research in machine learning and computer vision over the last year. In this time, Jacob has proven himself to be an exceptionally talented and productive researcher, with technical abilities far beyond those typically expected. I have advised many students on research projects, and Jacob is undoubtedly among the top X% in terms of initiative, coding skill, and understanding of advanced machine learning concepts. I offer my strongest recommendation that he be awarded the NSF fellowship.

I first met Jacob when he sought out opportunities to apply machine learning to his radar object detection project under Dr. Yi Wang. Despite his primary background in radar, he quickly impressed me with his command of computer science and eagerness to delve into complex machine learning methods. Jacob has made immediate, impactful contributions. From an intellectual merit standpoint, Jacob's work has been outstanding. He tackled some of our most challenging projects, demonstrating both deep theoretical knowledge and practical software engineering acumen. Notably, Jacob contributed to multiple facets of our projects' machine learning research.

Jacob independently implemented a state-of-the-art Masked Autoencoder (MAE) for temporal data – essentially extending the original image-based MAE paradigm to work with sequences of radar and camera frames. This Temporal MAE project required reading and synthesizing recent research papers, writing complex deep learning code, and a great deal of experiment tuning. Jacob handled this with skill and maturity, producing a working model that he is now using to analyze time-series sensor data.

While training large neural networks, Jacob identified memory and speed bottlenecks on our GPU clusters. He performed an in-depth analysis of GPU sharding and distributed training techniques to split neural network models across multiple GPUs. Jacob not only reviewed the latest parallel computing strategies but also integrated a multi-GPU training approach into the pipeline. This kind of systems-level optimization is typically the domain of experienced graduate researchers or software engineers, yet Jacob executed it flawlessly.

Throughout all these projects, Jacob brought a level of software professionalism that is rare in academic settings. He built modular, reusable libraries for sensor data processing and neural network training specific to our research needs. His code is clean,

well-documented, and even includes unit tests, reflecting an engineering discipline beyond his years. I have had students whose coding skills do not match Jacob's in clarity and reliability. This strong foundation will serve him extremely well in graduate research, where complex projects require maintainable codebases.

One aspect of Jacob's work that I particularly want to highlight is how he connects his machine learning efforts to tangible research outcomes and future directions. Jacob seeks to understand and improve the entire research cycle, not just experimentation. For example, after implementing the Temporal MAE model, he went a step further to analyze how that unsupervised pre-training could improve downstream task performance for detecting small objects in our radar-camera fusion project. This kind of big-picture thinking – linking low-level technical details to high-level research questions – is a hallmark of intellectual merit. Clearly, Jacob has the mindset of a scholar who can both innovate and communicate.

Equally important are Jacob's contributions that speak to the Broader Impacts criterion. He has a strong ethos of sharing knowledge and enabling others, which I have observed repeatedly. He has expressed a clear intention to open-source the radar-camera fusion dataset that he's helping to assemble as part of his graduate research plan. By making this data publicly available, Jacob will enable other researchers to advance the state of maritime object detection – an outstanding broader impact that aligns perfectly with NSF's mission to expand scientific knowledge for society's benefit. It is refreshing to see a young researcher so committed to collaboration and community advancement.

I have absolute confidence in Jacob's potential to become an influential researcher and educator in the future. He will thrive in a top graduate program, and his work – particularly in the proposed integration of AI with radar sensing for maritime safety – has promise to yield both scientific breakthroughs and societal benefits. I give him my highest recommendation for the NSF GRFP. Jacob has my full support, and I believe he fully exemplifies the qualities NSF seeks in its fellows.

Sincerely,
Dr. Yan Tong
Professor, Department of Computer Science & Engineering
University of South Carolina